

AP CALCULUS AB – UNIT 5

5.2–5.5 II Derivative Graph Analysis
- Part 2: Challenge Questions

Name: _____ Class: _____ Date: _____

Lesson Overview and Learning Objectives

- Determine intervals of concavity and locate points of inflection.
- Use the Second Derivative Test when its hypotheses are satisfied.
- Relate the graphs and signs of f , f' , and f'' .
- Synthesize first- and second-derivative information in complete curve analysis.

Keys: Essential Knowledge and Vocabulary

Concave up/down	$f'' > 0$ means slopes f' increase; $f'' < 0$ means slopes decrease.
Inflection point	A point on the graph where concavity changes; $f'' = 0$ alone is only a candidate.
Second Derivative Test	If $f'(c) = 0$, then $f''(c) > 0$ gives a local min and $f''(c) < 0$ gives a local max; $f''(c) = 0$ is inconclusive.
Graph of f'	Where f' increases, f is concave up; where f' decreases, f is concave down.
Graph of f''	Its sign directly determines concavity of f and increase/decrease of f' .
Complete analysis	Combine domain, intercepts, asymptotes, critical points, monotonicity, concavity, and end behavior.

Explanation, Strategy, and Common Errors

- Solve $f'' = 0$ and include points where f'' is undefined, then test concavity on the resulting intervals.
- Verify an actual sign change in concavity before naming an inflection point.
- When reading f' , distinguish its sign (monotonicity of f) from its slope (concavity of f).
- The Second Derivative Test applies only at a critical point with a suitable second derivative sign.
- A complete AP explanation links the derivative evidence to a precise conclusion about the original function.

Solved Example: Concavity

For $f(x) = x^4 - 4x^3$, determine concavity and inflection points.

Answer and Reasoning

$f'' = 12x(x - 2)$. Concave up on $(-\infty, 0) \cup (2, \infty)$ and down on $(0, 2)$. Inflection points at $x = 0$ and $x = 2$, with points $(0, 0)$ and $(2, -16)$.

Solved Example: Second Derivative Test

Classify critical points of $f(x) = x^3 - 3x^2 + 2$ using f'' .

Answer and Reasoning

$f' = 3x(x - 2)$ gives $x = 0, 2$; $f'' = 6x - 6$. $f''(0) = -6$, local max; $f''(2) = 6$, local min.

Solved Example: From the graph of f'

If f' is positive and decreasing on $(1, 4)$, describe f .

Answer and Reasoning

f is increasing because $f' > 0$ and concave down because f' is decreasing.

Solved Example: Full sign synthesis

Using the shared table, describe the behavior at $x = -2$ and $x = 1$.

Answer and Reasoning

The f' sign changes $+$ $\rightarrow$ $-$ at -2 , giving a local maximum. It changes $-$ $\rightarrow$ $+$ at 1 , giving a local minimum. Concavity information must be read from the f'' row on neighboring intervals.

Shared Representation / Data

Interval	$(-\infty, -2)$	-2	$(-2, 1)$	1	$(1, \infty)$
$f'(x)$	+	0	−	0	+
$f''(x)$	−		− then +		+

AP Exam Language

For theorem applications, state every hypothesis. For graph analysis, justify sign changes and compare all required candidates. For optimization, state the objective, constraint, feasible domain, and verification.

Leveled Practice**Easy Foundations**

Build fluency with the definition, notation, and one-step applications.

1. If $f'' > 0$ on an interval, describe concavity.

2. If f' is decreasing, what is the sign of f'' ?

3. Does $f''(c) = 0$ alone prove an inflection point?

4. If $f'(c) = 0$ and $f''(c) < 0$, classify c .

Medium Applications

Connect representations and apply the lesson procedure in familiar settings.

5. For $f(x) = x^3 - 6x$, find the inflection point.

6. For $f''(x) = (x - 2)(x + 1)$, determine where f is concave down.

7. If f' has a local maximum at $x = 3$, what happens to concavity of f there?

8. At a critical point c , $f''(c) = 0$. What test should be used next?

Hard Reasoning

Select a method, justify conclusions, and combine several ideas.

9. For $f(x) = x^4 - 8x^2$, find local extrema and inflection points.

10. If $f' < 0$ and $f'' > 0$, describe f .

11. A graph of f'' crosses upward through zero at $x = a$. What can be said about f' and f ?

12. Explain why a vertical tangent can coincide with an inflection point.

Difficult Analysis

Work through less-direct algebra, subtle hypotheses, and multi-step reasoning.

13. For $f'(x) = x^2 e^{-x}$, determine monotonicity and concavity of f .

14. Suppose f'' is positive except $f''(2) = 0$. Is $x = 2$ an inflection point?

15. A function has $f'(x) = (x - 1)^3(x + 2)$. Analyze local extrema and concavity candidates.

Challenge / AP Extension

Synthesize the topic in unfamiliar, proof-oriented, or parameter-based problems.

16. Construct a function with $f''(0) = 0$ but no inflection at 0.

17. Construct a function with an inflection at 0 where $f''(0)$ does not exist.

AP-Style Multiple-Choice Practice

Part A: No Calculator

Choose the best answer. Justification is not required on the AP multiple-choice section, but show reasoning while practicing.

1. An inflection point requires

- (A) $f'' = 0$ only
- (B) a local extremum
- (C) a change in concavity
- (D) $f' = 0$

2. If $f' > 0$ and $f'' < 0$, then f is

- (A) increasing and concave down
- (B) decreasing and concave down
- (C) increasing and concave up
- (D) decreasing and concave up

3. If $f'(c) = 0$ and $f''(c) > 0$, then

- (A) local maximum
- (B) local minimum
- (C) inflection necessarily
- (D) no conclusion

4. Where the graph of f' is increasing, f is

- (A) decreasing
- (B) concave up
- (C) concave down
- (D) constant

Part B: Graphing Calculator Permitted

Use a calculator only for numerical evaluation, graphing, or regression as indicated. Preserve mathematical setup.

5. For $f(x) = x^4 - 4x^2$, an inflection input is closest to

- (A) 0
- (B) 0.58
- (C) 0.82
- (D) 1.41

6. If $f'(x) = xe^{-x}$, f is concave down for

- (A) $x < 0$
- (B) $0 < x < 1$
- (C) $x > 1$
- (D) all x

AP-Style Free-Response Practice

No Calculator

Let $f(x) = x^4 - 4x^3 + 2$.

(a) Find critical points and classify them. [3 points]

(b) Determine concavity and inflection points. [3 points]

(c) Explain the behavior at $x = 0$ using both derivatives. [2 points]

Calculator Permitted

A calculator graph of f' has zeros at $x = -1.3$ and $x = 2.4$, a local maximum of f' at $x = 0.5$, and $f' > 0$ outside the two zeros.

(a) Classify the two critical points of f . [2 points]

(b) Identify the inflection input of f indicated by the local maximum of f' . [2 points]

(c) Describe f on $(-1.3, 0.5)$ and $(0.5, 2.4)$. [2 points]